

7 Gel electrophoresis is used to separate DNA fragments of different lengths.

The rate at which the DNA fragments move through the gel depends on several factors including:

- molecular size of the DNA fragment
- shape of the DNA fragment
- concentration of the gel.

(a) (i) Which enzyme is used to cut the DNA into fragments?

(1)

- ☐ **A** DNA polymerase
- ☐ **B** integrase
- ☐ **C** restriction enzyme
- ☐ **D** reverse transcriptase

(ii) Explain why the use of an enzyme to cut the DNA results in fragments, of different lengths, that can be separated by gel electrophoresis.

(3)

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(b) Fragments of double-stranded DNA move through the gel at a relative rate (M_r) that is inversely proportional to the log of their molecular weight (MW).

(i) Complete the table using the equation:

$$M_r = \frac{1}{\log_{10} MW}$$

(2)

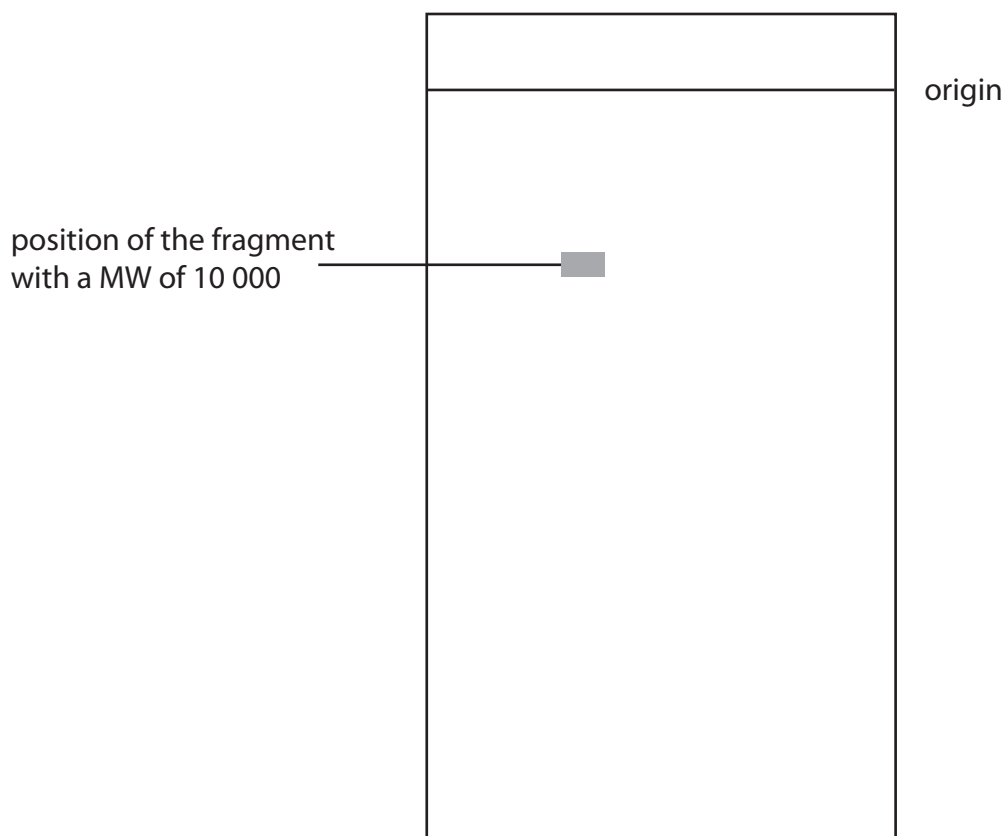
Molecular weight of DNA fragment (MW)	Relative rate of movement (M_r)
100 000	
10 000	0.25
	0.34

(ii) The diagram shows the position of a DNA fragment with a MW of 10 000, after gel electrophoresis.

Complete the diagram to show the position of a DNA fragment with a MW of 100 000.

Use the information in the question.

(1)



(c) The fragments move more slowly through a higher concentration of gel.

Suggest why the fragments move more slowly through a higher concentration of gel.

(1)

(d) Circular DNA moves at a faster rate through the gel than linear DNA.

(i) Give **two** examples of circular DNA found in cells.

(2)

(ii) Give **two** differences between the structure of circular DNA and that of linear DNA, other than their shapes.

(2)

(Total for Question 7 = 12 marks)